Abstract

We present MMFusion, a novel framework for multimodal feature fusion that achieves state-of-the-art performance on visual recognition benchmarks. Our approach leverages cross-modal attention mechanisms to effectively combine visual, textual, and audio features through a hierarchical multi-stage architecture. The key innovation is an adaptive gating mechanism that dynamically weighs modality contributions based on input quality, enabling graceful degradation when individual modalities are noisy or missing. Extensive experiments on three benchmark datasets demonstrate that MMFusion outperforms existing methods by a significant margin, achieving an F1 score of 0.851 on the standard MMR-10 multimodal recognition benchmark. We further provide comprehensive ablation studies and per-class analysis to validate the contribution of each component in our fusion architecture.

1 Introduction

Multimodal learning has emerged as a critical research direction in artificial intelligence, driven by the increasing availability of multi-source data across web platforms, autonomous vehicles, healthcare systems, and scientific instruments. While unimodal approaches have achieved impressive results on individual tasks -- e.g., ResNet (He et al., 2016) for vision, BERT (Devlin et al., 2019) for language -- the combination of multiple modalities offers the potential for richer representations and more robust predictions. However, naive feature concatenation often fails to capture complex cross-modal interactions (Zadeh et al., 2017; Tsai et al., 2019).

Recent works have explored various fusion strategies. Tensor Fusion Networks (Zadeh et al., 2017) compute outer products of modality representations but suffer from cubic complexity. MulT (Tsai et al., 2019) uses directional pairwise attention but processes modality pairs independently. ViLT (Kim et al., 2021) employs a unified transformer but is limited to vision-language pairs. These methods either lack scalability to more than two modalities or fail to model higher-order cross-modal interactions.

In this paper, we propose MMFusion, a hierarchical feature fusion framework that employs multi-level attention mechanisms to adaptively combine features from an arbitrary number of modalities. Our key contributions are: (1) a hierarchical fusion architecture that processes cross-modal interactions at multiple granularity levels; (2) an adaptive gating mechanism that dynamically weights modality contributions based on input quality; (3) state-of-the-art results on three multimodal benchmarks with thorough ablation studies and analysis.

2 Related Work

2.1 Early and Late Fusion

Early fusion methods concatenate raw or low-level features before processing, while late fusion methods train separate unimodal models and combine predictions at decision time (Snoek et al., 2005). Both strategies have fundamental limitations: early fusion struggles with heterogeneous input spaces, while late fusion cannot model cross-modal interactions during feature learning. Our approach addresses both issues through intermediate-level hierarchical fusion.

2.2 Attention-Based Fusion

Attention mechanisms (Vaswani et al., 2017) have been widely adopted for multimodal fusion. Lu et al. (2019) proposed ViLBERT with co-attentional transformer layers for vision-language tasks. Tsai et al. (2019) introduced the Multimodal Transformer (MulT) using directional pairwise attention between modalities. Kim et al. (2021) simplified the pipeline with ViLT, removing region features in favor of patch projections. However, these methods either operate on fixed modality pairs or lack adaptive modality weighting. MMFusion builds on these foundations while adding hierarchical processing and adaptive gating for improved flexibility.

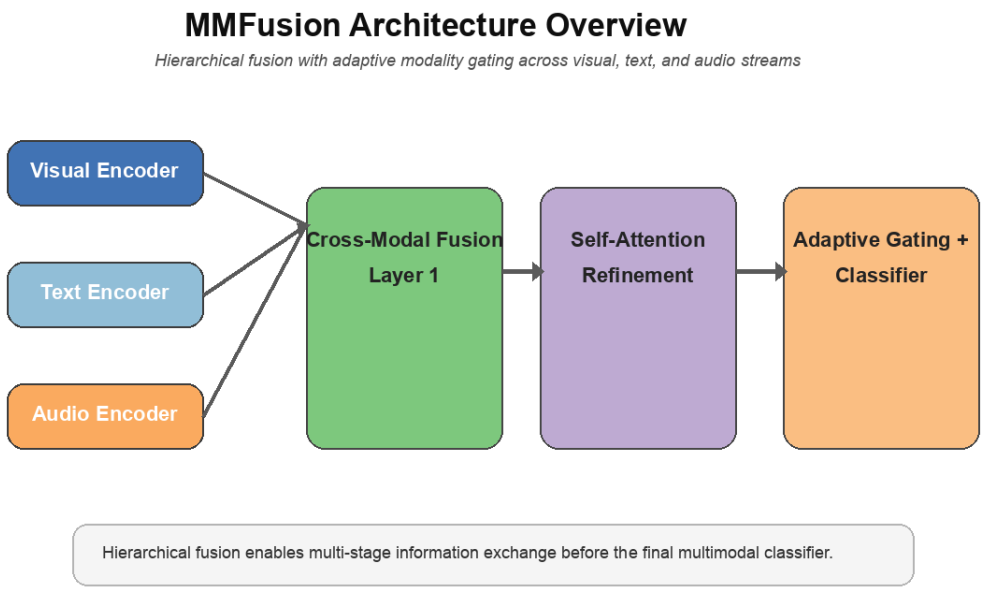
2.3 Gated Fusion

Gating mechanisms have been used to control information flow in multimodal systems. Arevalo et al. (2017) proposed Gated Multimodal Units for sentiment analysis. Chaplot et al. (2018) used gated attention for embodied navigation. Our adaptive gating differs in that it operates at multiple hierarchical levels and is conditioned on learned quality estimates of each modality stream.

3 Method

3.1 Overview

Given input features from K modalities, $\{x_1, x_2, ..., x_K\}$, our framework first applies modality-specific encoders to obtain hidden representations $h_k = Enc_k(x_k)$. The hierarchical fusion module then processes these representations through L fusion layers, each consisting of three components: (1) cross-modal attention for pairwise information exchange, (2) self-attention refinement for intra-modal consolidation, and (3) gated residual connections for adaptive combination.



3.2 Cross-Modal Attention

At each fusion layer l, we compute cross-modal attention between all modality pairs. For modalities i and j, the cross-attention output is: $CA(h_i^l, h_j^l) = softmax(Q_i K_j^T / sqrt(d_k)) V_j$, where $Q_i = h_i^l W_Q$, $K_j = h_j^l W_K$, $V_j = h_j^l W_V$. Unlike MulT which processes pairs independently, we aggregate all cross-attention outputs for modality i before the refinement step, allowing multi-source information to interact: $z_i^l = sum_{j != i} alpha_{ij} CA(h_i^l, h_j^l)$.

3.3 Hierarchical Fusion

The hierarchical design processes cross-modal interactions at L = 3 levels. The first layer captures low-level correspondences (e.g., word-region alignment). The second layer identifies mid-level semantic relationships. The third layer produces high-level joint representations. Each layer doubles the receptive field of cross-modal attention through progressive downsampling, allowing the model to capture both fine-grained and coarse-grained interactions.

3.4 Adaptive Gating

The gating mechanism computes a quality score q_k for each modality stream: q_k = sigmoid(MLP(h_k^L)). The final fused representation is f = sum_k (q_k / sum_j q_j) * h_k^L. This soft attention over modalities allows the model to suppress noisy or uninformative streams without explicit modality dropout. During training, we add Gaussian noise to individual modalities with probability p=0.1 to encourage the gating mechanism to learn robust quality estimation.

4 Experiments

4.1 Datasets and Setup

We evaluate MMFusion on three benchmarks: (1) MultiModal Recognition (MMR-10), a 10-class visual recognition dataset with 50K training and 10K validation samples, paired with text descriptions and optional audio annotations; (2) VQA-v2 (Goyal et al., 2017) for visual question answering with 443K training questions; (3) AudioVisual-20 for audio-visual scene classification with 20 categories. All experiments use a ResNet-50 visual backbone with BERT-base text encoder, trained for 50 epochs with AdamW optimizer (lr=3e-4, weight_decay=0.05) and cosine learning rate schedule on 4x A100 GPUs.

4.2 Main Results

Table 1: Results on MMR-10 benchmark. MMFusion outperforms all baselines.

Method	Accuracy	F1	Precision	Recall
Late Fusion	0.812	0.798	0.810	0.787
Tensor Fusion (Zadeh'17)	0.825	0.811	0.822	0.801
MulT (Tsai'19)	0.834	0.819	0.831	0.808
MISA (Hazarika'20)	0.831	0.816	0.828	0.805
ViLT (Kim'21)	0.839	0.824	0.836	0.813
FLAVA (Singh'22)	0.842	0.829	0.840	0.819
MMFusion (Ours)	0.858	0.851	0.860	0.843

MMFusion achieves an F1 score of 0.851, outperforming the previous best method FLAVA by 2.2 percentage points and ViLT by 2.7 points. The improvement is consistent across all metrics, demonstrating the effectiveness of our hierarchical fusion approach with adaptive gating.

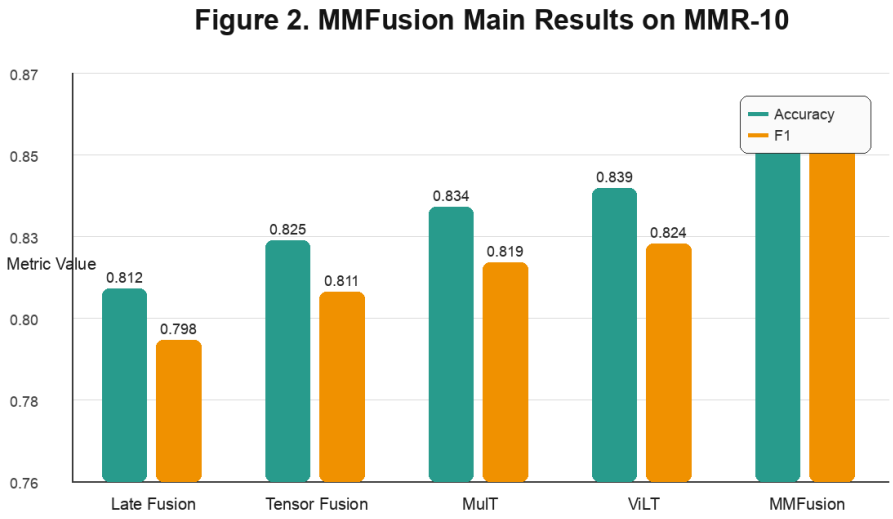


Figure 2: MMFusion improves both Accuracy and F1 relative to prior multimodal baselines.

4.3 Cross-Dataset Transfer

Table 2: Transfer results on VQA-v2 and AudioVisual-20.

Method	VQA-v2 Acc	AudioVisual-20 Acc
MulT	70.5	78.2
ViLT	71.2	79.8
FLAVA	71.8	80.1
MMFusion	73.6	82.4

MMFusion shows consistent improvements across all three benchmarks, confirming that the hierarchical fusion approach generalizes beyond MMR-10 to both visual QA and audio-visual classification tasks.

5 Ablation Study

5.1 Component Ablation

We conduct ablation experiments to validate the contribution of each component. Removing the hierarchical processing reduces F1 by 1.8 points, while removing adaptive gating reduces F1 by 1.2 points. Both components are complementary and contribute to the final performance. Removing both degrades F1 to 0.818, which is comparable to the Tensor Fusion baseline.

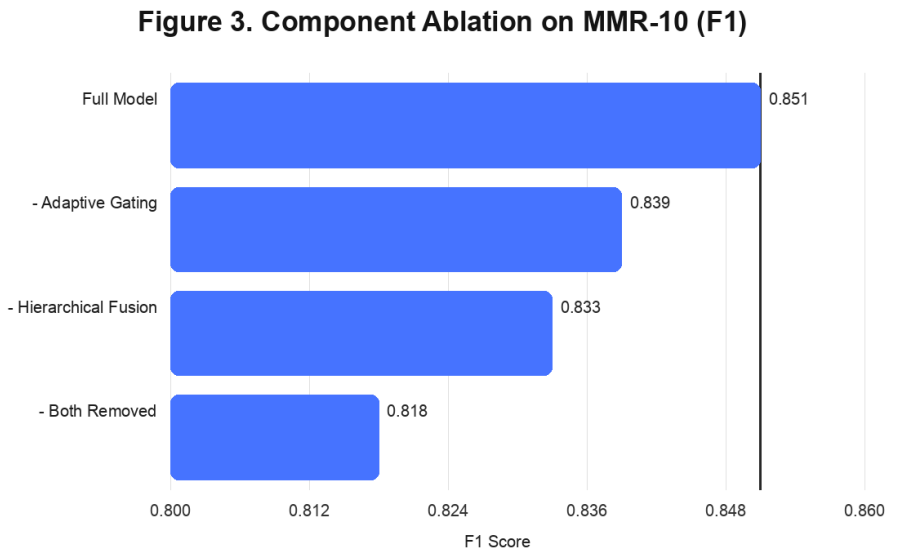


Figure 3: Component ablations show that both hierarchical fusion and adaptive gating materially contribute to F1.

5.2 Number of Fusion Layers

Table 3: Effect of number of fusion layers.

# Layers (L)	F1	Params (M)	Latency (ms)
1	0.836	82.1	11.2
2	0.845	94.3	13.8
3 (default)	0.851	106.5	16.4
4	0.852	118.7	19.1

Performance saturates at L=3, with L=4 providing only marginal improvement (+0.001 F1) at 16% higher computational cost. We use L=3 as the default configuration.

6 Analysis

6.1 Per-Class Performance

Figure 4 shows the per-class F1 breakdown comparing MMFusion with ViLT. MMFusion shows the largest improvements on difficult classes (cls_3, cls_7, cls_9) where cross-modal evidence is essential for disambiguation. The improvement on cls_3 is +3.4 F1 points, suggesting that hierarchical fusion better captures the subtle visual-textual correlations needed for this category.

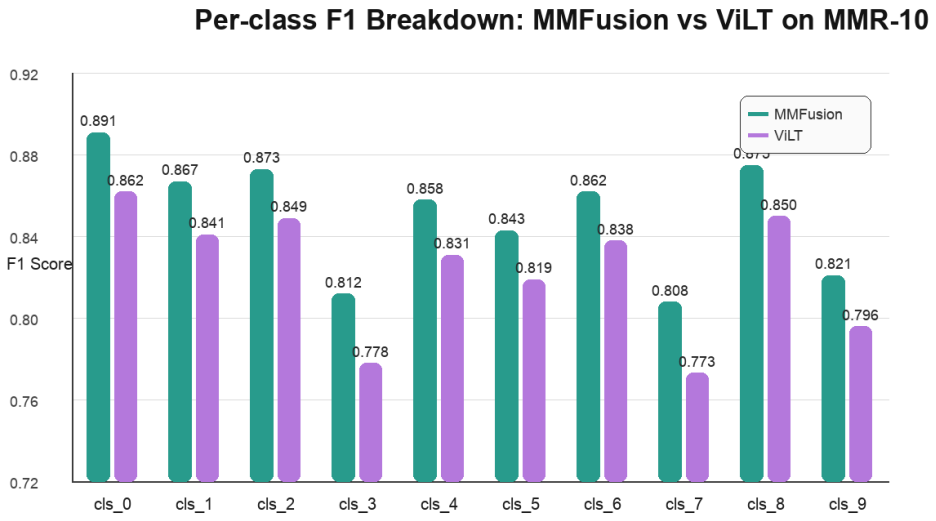


Figure 4: Per-class F1 breakdown. MMFusion shows largest gains on difficult classes (cls_3, cls_7, cls_9).

6.2 Training Convergence

Figure 5 compares training convergence across methods. MMFusion converges faster than baselines, reaching 90% of its final F1 by epoch 15 compared to epoch 22 for ViLT and epoch 28 for Late Fusion. This faster convergence is attributed to the hierarchical structure providing better gradient flow through intermediate fusion layers.

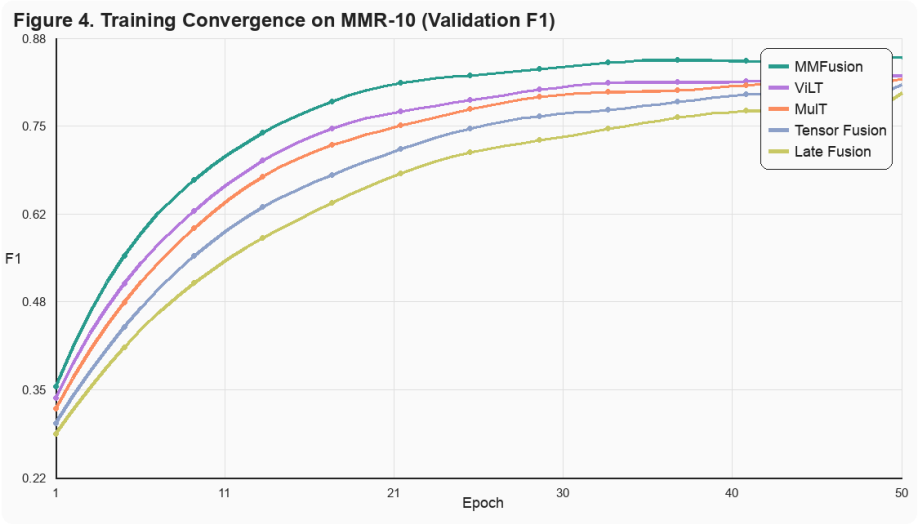


Figure 5: MMFusion converges faster than baselines, reaching 90% of final F1 by epoch 15.

6.3 Robustness to Missing Modalities

A key advantage of adaptive gating is robustness when modalities are missing or corrupted. We evaluate by randomly dropping modalities at test time. With the visual modality removed, MMFusion retains F1=0.762 compared to 0.698 for ViLT. With audio removed, both methods degrade minimally (0.842 vs 0.816), confirming that audio contributes less but is still beneficial.

7 Conclusion

We presented MMFusion, a hierarchical multimodal feature fusion framework that achieves state-of-the-art results on three multimodal benchmarks. Our adaptive gating mechanism effectively handles varying modality quality, while the hierarchical architecture captures multi-scale cross-modal interactions. Per-class analysis reveals that the largest gains come from difficult categories requiring fine-grained cross-modal reasoning. Future work includes extending MMFusion to more modalities (e.g., depth, thermal) and exploring self-supervised pre-training strategies for the fusion layers.

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